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Reflection: burden of cervical cancer in Sub-Saharan Africa and progress with HPV vaccination

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In Sub-Saharan (SSA) cervical cancer is the leading cause of cancer deaths amongst women. The region carries the greatest burden, with 24.55% of the global mortality from cervical cancer. Reports indicate an increasing challenge of cervical cancer in SSA.

HPV vaccination with its well-established effectiveness provides hope for cancer control in SSA. Following an initial delay in HPV vaccine uptake in SSA, 18 countries mostly in Eastern and Southern Africa, had a national programme by 2020. Vaccination coverage data show that high populated countries have lower coverage figures. Furthermore, high coverage of demonstration projects may not be achieved in the national rollout.

In conclusion, whilst there is significant progress with the rollout of HPV vaccination programme in SSA, some countries in West Africa should be prioritised. Experiences of early adopters should be reviewed to guide other countries to achieve and sustain high coverage.

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Introduction

Cervical cancer is a significant global public health problem that can be prevented. Globally an estimated 311 365 women succumbed to the disease in 2018 [1•]. It is projected that at the current level of intervention, the global burden of cervical cancer will increase [2•]. The World Health Organization (WHO) highlights that eliminating cervical cancer will contribute to four sustainable development goals (SDGs): poverty alleviation; good health and wellbeing; reduction of gender inequalities; and reduction of inequalities. In 2018 WHO Director General made a 'Call to Action' for cervical cancer elimination [3].

Human papillomavirus (HPV) is the necessary cause of cervical cancer. Persistent infection with high risk types of HPV activate the progression of a normal cell into precancerous lesions leading to cervical cancer [4,5]. HPV types 16 and 18 account for about 70% of cervical cancer cases globally [6]. Currently, three types of HPV vaccines are available: bivalent and quadrivalent available since 2006/7 and the nonavalent (9 valent) vaccine licensed in 2014 by the Food Drug and Administration of the USA [7–9].

HPV vaccines are highly effective in preventing precancerous cervical lesions caused by vaccine-specific types [7,8]. Furthermore, studies have demonstrated population impact indicating herd immunity from HPV vaccines. Drolet *et al.* [10•] reported from a systematic review and meta-analysis, a decrease of cervical intraepithelial neoplasia grade 2 (CIN2) in 15–19 and 20–24 year olds by 51% and 31%, respectively [10•]. After 5–8 years of HPV vaccination, type 16 and 18 infections decreased by 83% and by 66% in 13–19 and 20–24 year olds, respectively. The study reported cross-protection and herd protection where coverage was 50% or higher [10•].

CIN grade 2 and 3 lesions are monitored as the proximal outcome for cervical cancer because it will take decades before the full impact of HPV vaccination is realised. Even so, preliminary data and observations from cancer registries have shown a decrease in cervical cancer and later stages of CIN. These include a report by Luostarinen *et al.* [11] on follow-up on women recruited for PATRICIA and FUTURE studies and observation on the Finnish cancer registry with comparison with unvaccinated women. Other reports also show similar results and show herd effect and cross protection [11–14].

This paper reviewed the burden of cervical cancer in Sub-Saharan Africa (SSA) in relation to progress made with HPV vaccination in the context of the elimination goal and suggest areas of focus on way forward with the HPV vaccination programme in SSA.

Increasing burden and trends of cervical cancer in Sub-Saharan Africa

Cervical cancer best demonstrates the impact of poverty on health. Sengayi-Muchengeti *et al.* [15•] conducted an analysis on association of cervical cancer incidence and mortality with the human development index (HDI). It shows that populations with low HDI experience the highest incidence and the worst outcomes [15•]. Of the global population of women aged 15 years and older, who are at risk of cervical cancer, 10.2% live in SSA. However, SSA accounts for 19.59% (111 632) and 24.55% (76 444) of the global burden of cases and deaths per year, respectively [2•,10•]. The incidence rates of cervical cancer in some countries in SSA are the highest in the world. Swaziland and Malawi, have the highest incidence in Africa, with age standardised incidence rates (ASR) of 75.3 and 72.9 per 100 000 women, respectively.

Cervical cancer challenges for SSA are not about to dissipate. Jedy-Agba *et al.* [16•] reported an increase in incidence rate of cervical cancer in SSA, based on an analysis of 10 population cancer registries of eight countries covering Western, Eastern, and Southern Africa. The largest increase was in Malawi and Eastern Cape in South Africa, at an annual increase of 7.9% and 3.5%, respectively [16•]. Cervical cancer in SSA affects younger women in the reproductive age, at the prime of their lives and is diagnosed in a much younger age group [1•,17].

An additional burden is the region's highest HIV/AIDS prevalence in the world. Adolescent girls and young women bear the greatest burden of the disease and have eight times higher risk of HIV than their male counterparts [18]. UNAIDS [18] highlights the synergy between HPV and HIV infections in young women in SSA. HIV infected women are at four to six times higher risk of cervical cancer than uninfected women [2•,18].

Other challenges for SSA is the poor performance of screening programmes, inconsistent and poorly implemented screening policies [1•,18,19]. PAP screening coverage ranges from 2% to 20% in urban areas and from <1% to 14% in rural areas [19]. Screening rates for: Kenya, Malawi, Mozambique, and South Africa are 3.2%, 2.6%, 1% and 19.3%, respectively [1•]. It is not surprising that the five year survival of women with cervical cancer in SSA is 33%, much lower than the 80% of high-income countries (HICs) countries [15•].

The death of young women impacts on their families, specifically their children. Vega *et al.* [17] estimated that in Africa for 100 women who die from breast and cervical cancer, 14 to 30 children die as an indirect result of their mother's death, and child mortality increases by up to nine times as a direct cause [17]. This impact was higher in countries with high child mortality and higher ASR of cervical cancer such as: Mozambique, Nigeria, Somalia and Sierra Leone.

Elimination of cervical cancer and progress with HPV vaccination in SSA

The Global Strategy to Accelerate the Elimination of Cervical Cancer as a Public Health Problem has set 4 cases per 100 000 women as an elimination threshold [2•].The 90–70–90 targets to be met by 2030 for countries to be on the path towards elimination are:

- 90% of girls fully vaccinated with the HPV vaccine by age 15.
- 70% of women are screened with high quality tests by 35, and again by 45 years.
- 90% of women identified with cervical disease receive treatment.

WHO recommends a combination of interventions to achieve the set targets and that countries introduce HPV vaccine in national immunisation programmes targeting girls 9–14 years, before the age of 15; multiple cohorts in this age group be vaccinated when the vaccine is first introduced.

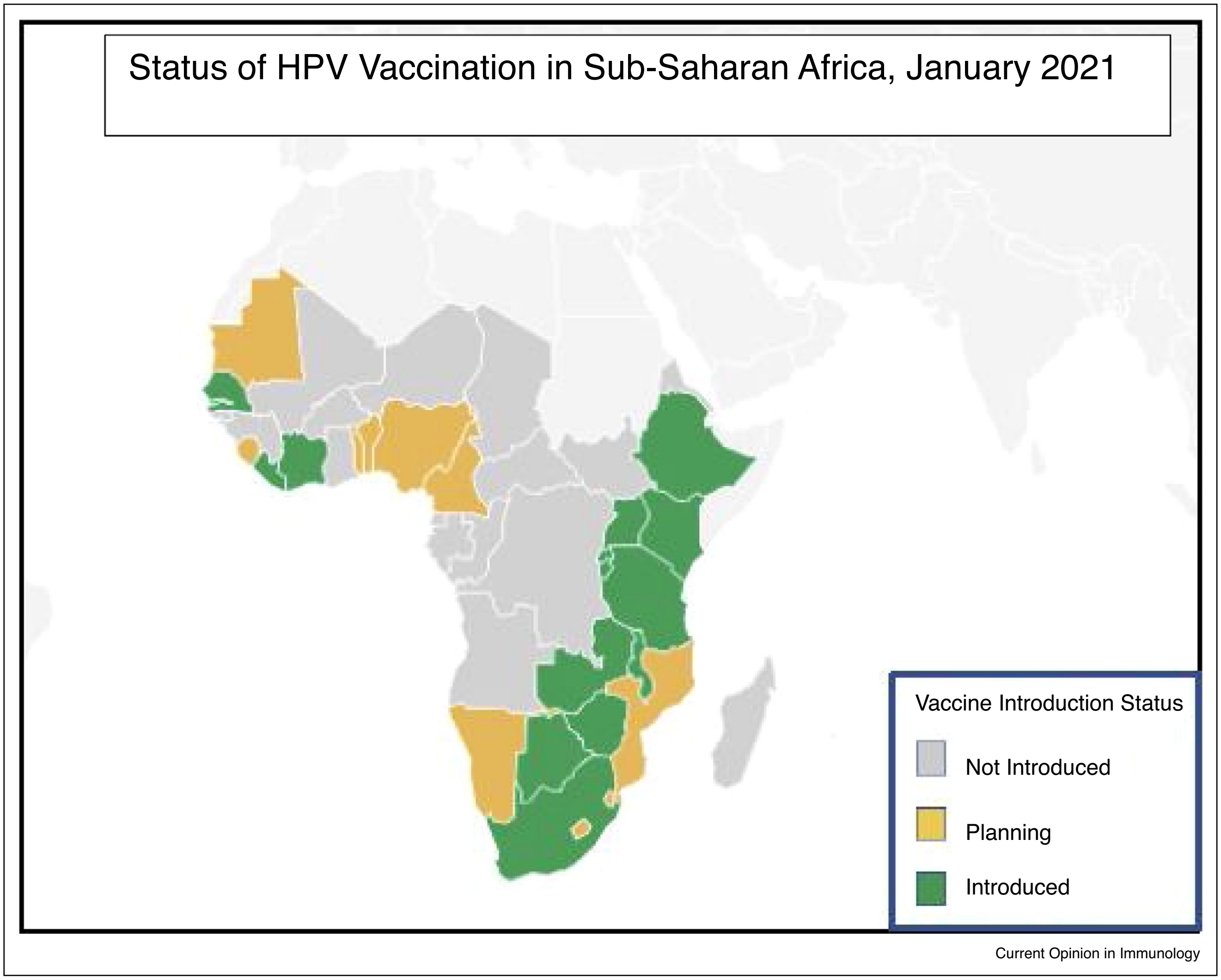
Abass *et al.* [20•], in a recent modelling and analysis that used revised demographic figures incorporating aging population shows that HPV vaccination provides more benefit and is more cost effective than previously thought. It demonstrates that Africa will obtain the greatest benefits from HPV vaccination; 28 cases, 23 deaths and 470 disability adjusted life years (DALYs) will be averted per 1000 girls vaccinated. The impact was higher where girls were vaccinated at 9 years than at 12 years.

Many low-income and middle-income countries (LMICs) conducted demonstration projects to assess feasibility of the HPV vaccination programme. Although these projects started early; 2008 in Uganda, 2010 South Africa and 2011 in Rwanda, the national rollout came later [21–24]. In 2010, no country in SSA had a national programme; in 2014, five countries had (Botswana, Lesotho, Rwanda, South Africa, and Seychelles). With Gavi support, eight countries early in 2018 (which covered a very small fraction of the population); additional three countries by end of 2018 and 18 countries in 2020. Figure 1 is a map of countries that have introduced the HPV vaccines as of January 2021 [25].

Since 2020, a much higher percentage of the eligible population will be covered. Additional countries with their ASR are: Ethiopia – 18.9; Kenya – 33.8; Malawi – 72.9; Tanzania – 59.1; Zambia – 66.4 and Zimbabwe – 62.3; are strategically important in cervical cancer control because of their very high incidence rates and large populations.

In SSA, South and Eastern have the highest ASR of 43.1 and 40 per 100 000 women; followed by Western

Figure 1



Status of HPV vaccine introduction in Sub-Saharan Africa.
Source: International Vaccination Centre, January 2021 [25].

and Middle Africa at 29 and 26, respectively. Ten of the 18 (56%) countries in Eastern Africa and two of the five in Southern Africa have rolled out the programme. This is in line with 41% (120.82/293.16 million) of women at risk in Eastern Africa and the 10 countries with the programmes have 76.56% the population at risk. Southern Africa is a special case because South Africa has over 90% of the population.

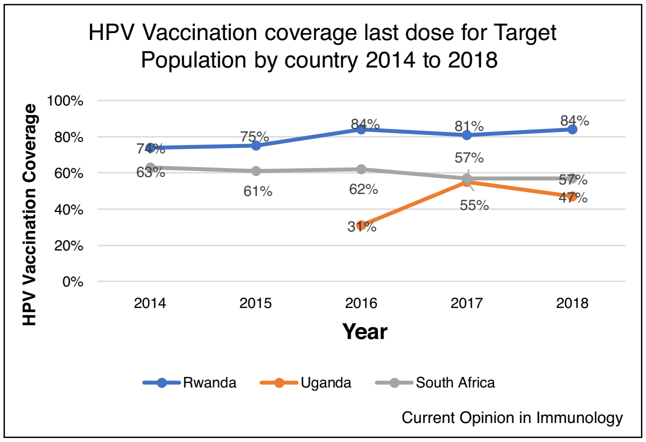
Four (25%) of 16 countries in Western Africa have a national vaccination programme, lower than the corresponding proportion of 35.7% (104.65/293.16 million) of women at risk. Middle Africa with 15.24% women at risk has one country with the programme.

HPV vaccine uptake is aligned with the ASR of cervical cancer per sub-region. However, there are country variations. In West Africa: Burkina Faso, Guinea, Mali and Ghana have high ASR of: 45.1; 45.5; 43.9 and 32.9, respectively. Furthermore, these countries have large populations with correspondingly many cases; 3151 in Ghana and 2517 in Burkina Faso per year based on 2018 estimates [19]. Yet national HPV vaccination programme has not been rolled out in these countries.

HPV vaccination coverage

According to Cervical Cancer Action, as of mid 2020, only 15% of the global targeted population had been vaccinated [22]. How much of the eligible group in SSA is

Figure 2



HPV vaccination coverage with last dose in targeted group for selected countries.
Data Source; WHO Coverage estimates, accessed Nov 2020.

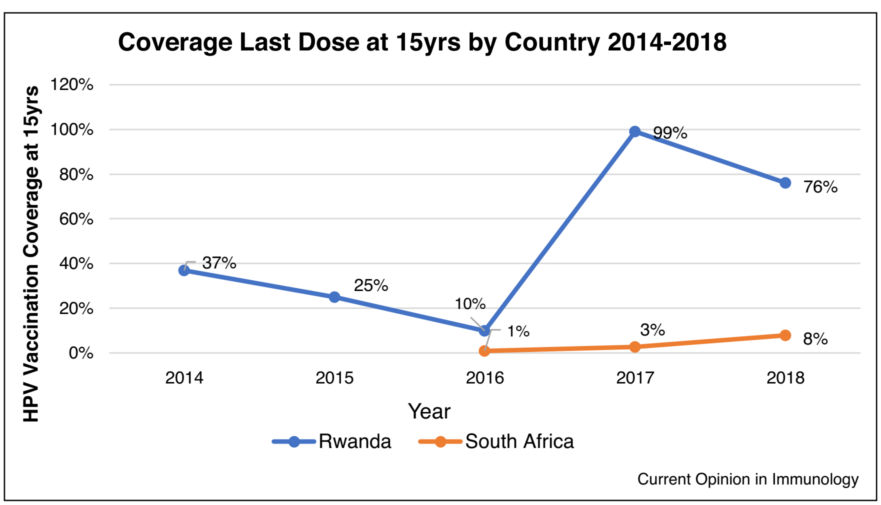
targeted and how is the programme performing? WHO HPV vaccination coverage data is reviewed for selected countries and presented in Figure 2. The selected countries had an HPV vaccination programme for four years up to 2018 and large populations. Of eight countries with data: Seychelles, Botswana, Lesotho (later stopped) and Sao Tome and Principe have high coverage but have very small populations. South Africa and Uganda have high populations but low coverage. Rwanda's coverage is high, but data quality is a challenge: coverage above 100% (adjusted to 99%); and full dose coverage of girls at 15 years in 2017 rose from 10% to over 100% in one year (Figure 3). Similar data quality concerns are noted for Uganda and South Africa.

Last dose coverage in South Africa is decreasing, from 63% to 57% for target population. In 2018, coverage for girls at 15 years in South Africa was 8%. Rwanda's coverage for girls at 15 years dropped from 99% in 2017 to 76% in 2018. The decrease in coverage can be expected to translate to a decrease in coverage at 15 years in the next couple of years. Low coverage at 15 years is a serious concern because this is an indicator of expected level of protection. The observation is: coverage attained in national rollout are much lower than coverage obtained during demonstration projects. High coverage figures, 70–90%, were reported in demonstration projects in LMICs: [23,24,26,27]. In spite of data quality concerns, the data provides an indication of programme performance

This observation of lower national coverage figures compared to demonstration projects is supported by reports from Uganda. A study in Uganda, on 10–14 year old girls who were eligible for vaccination found a low uptake of the vaccine, 22% of the girls were vaccinated and 78% were not. Furthermore, older girls (13 yrs) were more likely to be vaccinated than the younger ones [28]. Another study in Uganda conducted in 2016 in a district that piloted HPV vaccination, found that 50% of eligible girls were not vaccinated and 17,8% had completed all three doses [29•]. Uptake was significantly low, a serious concern since this district had conducted a demonstration project.

WHO coverage figures for South Africa are different from those reported elsewhere, of 93% and 73% for 1st and 2nd

Figure 3



Coverage with the last dose at 15 years for selected countries with data.
Data Source; WHO Coverage estimates, accessed Nov 2020.

dose in 2014. A drop-out from 1st dose to 2nd dose of 21% and 26 % in 2014 and 2016, respectively is reported [30,31]. Rwanda consistently reports high coverage [27,31]. Sayinzoga *et al.* [32•] report HPV coverage for cohorts turning 15 years in 2017 and 2018, was 99% and 83%, respectively. Discrepancy with WHO figures is acknowledged and the challenge of estimating coverage for birth cohorts is noted.

It is understandable why demonstration projects may not truly represent what a country will achieve with national rollout. Demonstrations are at a small scale, restricted to selected districts and some were selected for convenience [33]. Evidently, some challenges of national rollout may not be realised, or the large population and logistics of a national programme may not be fully appreciated. Challenges with sustaining initial achievements need investigation and close monitoring of performance is essential.

The selected countries are early adopters with over five years of HPV vaccination experience. They are expected to provide direction for those that follow with: high coverage; reliable coverage data; population and cancer registries that are linked to programme data to provide vaccination status of girls when they turn into young women and start cervical cancer screening.

For a majority of SSA countries the programme is school based, targeting a single cohort when first introduced [21,26,34–36]. Offering the vaccine in outreach services for out of school girls posed logistical challenges, requiring more resources, consequently very few countries offer this service [19]. Estimating the target population and coverage, especially for the birth cohort at 15 years is a challenge with a grade based model [32•,35].

The reality for the region is time constraints, less than a decade to implement the HPV vaccine and reach 90-70-90 goals by 2030. The impact of the ongoing COVID-19 pandemic on routine health services will compromise the programme.

Key considerations for SSA in moving forward towards elimination goal

- Implementation challenges that impact on the national rollout of programme should be investigated, mapped and addressed.
- Countries with low coverage should overhaul the programmes with evidence based locally relevant interventions to achieve and sustain high overage.
- Countries starting a national rollout to ensure that multiple cohorts are targeted and focus on younger girls, 9-year olds rather than 12 and older.
- Data quality and information systems should be addressed. HPV Vaccination Programme to be linked to population and cancer registries to allow a continuum of care for girls as they become women.
- Reliable coverage estimates are urgently required, surveys should be conducted; demographic health surveys to include HPV Vaccination coverage at 15 years.
- In line with UNAIDS' call, there should be an integrated approach to child, adolescent and reproductive health services.

Conclusion

There is significant progress with introduction of HPV vaccination in SSA. Experiences of early adopters should guide other countries, especially those with large populations. Efforts should improve information systems, ensuring that multiple cohorts are targeted and the national rollouts of the programme achieves and sustains high coverage figures.

Conflict of interest statement

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